

Data Structures Homework #2

資工二 113590021 邱冠勳

- Recall the **selection sort** we discussed in class to illustrate how to have the pseudo-code. The illustrated pseudo-code for selection sort is iterative as below.

ITERATIVE_SELECTION_SORT(A)

Input: An input array A of size n

Output: A sorted array // Sort the array $A[1 : n]$ in a nondecreasing order.

```

1  for  $i = 1$  to  $n$                                  $n+1$ 
2       $j = i$                                          $n$ 
3      for  $k = (i + 1)$  to  $n$                          $\frac{1}{2}(n-1) \cdot n + 2n$ 
4          if ( $A[k] < A[j]$ )                         $3 \cdot \frac{1}{2} \cdot (n-1) \cdot n$ 
5               $k \leftarrow j$   $\rightarrow 2 \cdot \frac{1}{2} \cdot (n-1) \cdot n$   $1 \cdot \frac{1}{2} \cdot (n-1) \cdot n$ 
6               $t = A[i]$                              $2n$ 
7               $A[i] = A[j]$                          $2n$ 
8               $A[j] = t$                              $2n$ 
9  return  $A$   $\rightarrow 2n$                                 $1$ 

```

- Count the total number of primitive operations used in ITERATIVE_SELECTION_SORT(A).

$$7 \cdot \left[\frac{1}{2} \cdot (n-1) \cdot n \right] + 12n + 2$$

$$= \frac{7}{2}n^2 - \frac{7}{2}n + 12n + 2$$

$$= \frac{7}{2}n^2 + \frac{17}{2}n + 2 = O(n^2)$$

Ans : $O(n^2)$

- With the same idea, please provide a recursive version for the selection sort, RECURSIVE_SELECTION_SORT(A), with pseudo-code.

{ Input : A is an unsorted array with n element, the sort will start from startIndex
 { Output : A sorted array // $A[\text{startIndex} : n]$ in ascending order

Function Recursive_Selection_Sort($A, m = 1$):

$n = \text{length}(A)$

If $m \geq n - 1$:

return A

$j = m$

for $k = (m + 1)$ to $n - 1$

if ($A[k] < A[j]$)

$j = k$

$t = A[m]$

$A[m] = A[j]$

$A[j] = t$

Recursive_Selection_Sort($A, m + 1$)

(c) Please try to derive the total number of primitive operations used in the recursive version you provided in 1b.

Function Recursive_Selection_Sort ($A, m=1$): $\rightarrow T(m)$

$n = \text{length}(A)$	2
If $m \geq n-1$:	2
return A	1
$j = m$	1
for $k = (m+1)$ to $n-1$	$(n-m) \cdot 2 + 2$
if ($A[k] < A[j]$)	$3 \cdot (n-m-1)$
$k++$ $j=k$	$1 \cdot (n-m-1)$
$t = A[m]$	2
$A[m] = A[j]$	2
$A[j] = t$	2
Recursive_Selection_Sort ($A, m+1$)	$1 + T(m+1)$

$$T(m) = T(m+1) + \{6 + 4 \cdot (n-m-1) + 2 \cdot (n-m)\}$$

$$= T(m+1) + 12 + 6(n-m)$$

$$= T(m+2) + 12 + 6(n-m-1) + 12 + 6(n-m)$$

$$= 5 + \sum_{m=1}^{n-1} [12 + 6(n-m)]$$

$$= 5 + 12(n-1) + 6 \cdot \sum_{m=1}^{n-1} (n-m)$$

$$= 5 + 12(n-1) + 6 \cdot \frac{n(n-1)}{2}$$

$$= 3n^2 + 9n - 7 = O(n^2)$$

Ans: $O(n^2)$

2. Show that for any real constants a and b , where $b > 0$, $(n+a)^b = \Theta(n^b)$.

$$f(n) = \Theta(g(n)) \rightarrow c_1 g(n) \leq f(n) \leq c_2 g(n) \quad f(n) = (n+a)^b \quad g(n) = n^b$$

$$(n+a)^b = n^b \cdot \left(1 + \frac{a}{n}\right)^b$$

$$\text{當 } n \gg a \text{ 時 } \left|\frac{a}{n}\right| \leq \frac{1}{2} \approx 2a \leq n$$

$$-\frac{1}{2} \leq \frac{a}{n} \leq \frac{1}{2}$$

$$\frac{1}{2} \leq \frac{a}{n+1} \leq \frac{3}{2} \quad \dots \text{同} +1$$

$$\left(\frac{1}{2}\right)^b \leq \left(\frac{a}{n+1}\right)^b \leq \left(\frac{3}{2}\right)^b \quad \dots \text{同時 } b \text{ 次方}$$

$$n^b \cdot \left(\frac{1}{2}\right)^b \leq n^b \cdot \left(\frac{a}{n+1}\right)^b \leq n^b \cdot \left(\frac{3}{2}\right)^b \quad \dots \text{同} \times n^b$$

$$n^b \left(\frac{1}{2}\right)^b \leq (n+a)^b \leq n^b \cdot \left(\frac{3}{2}\right)^b \quad \dots \text{得證 } (n+a)^b = \Theta(n^b) *$$

3. Consider the Fibonacci function $F(n)$, $F(1) = 1, F(2) = 2, F(3) = 3, F(4) = 5, \dots, F(n) = F(n-1) + F(n-2)$. Show by induction that $F(n)$ is $\Omega((3/2)^n)$.

(1) Base Cases :

$$\text{Let } c = \frac{2}{3}, n_0 = 1$$

$$F(1) = 1 = \Omega\left(\left(\frac{3}{2}\right)^1\right) \rightarrow 1 \geq \frac{2}{3} \times \frac{3}{2} = 1 \quad \dots \text{成立}$$

$$F(2) = 2 = \Omega\left(\left(\frac{3}{2}\right)^2\right) \rightarrow 2 \geq \frac{2}{3} \times \frac{9}{4} = \frac{3}{2} \quad \dots \text{成立}$$

(2) Inductive Hypothesis :

$$F(n-1) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-1}$$

$$F(n-2) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2}$$

(3) Inductive Step:

$$F(n) = F(n-1) + F(n-2)$$

$$F(n) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-1} + \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2}$$

$$F(n) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2} \cdot \left(\frac{3}{2} + 1\right) \quad \dots \text{提出 } \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2}$$

$$F(n) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2} \cdot \left(\frac{5}{2}\right)$$

$$F(n) = \Omega\left(\left(\frac{3}{2}\right)^n\right) \rightarrow F(n) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^n = \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2} \cdot \frac{9}{4}$$

$$\therefore \frac{5}{2} > \frac{9}{4}$$

$$\therefore F(n) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2} \cdot \left(\frac{5}{2}\right) \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^{n-2} \cdot \frac{9}{4} \geq \frac{2}{3} \cdot \left(\frac{3}{2}\right)^n = \Omega\left(\left(\frac{3}{2}\right)^n\right)$$

$$\text{ANS: 當 } c = \frac{2}{3}, F(n) = \Omega\left(\left(\frac{3}{2}\right)^n\right)$$

4. Let $f(n)$ be an asymptotically positive functions. Disprove with counterexamples or prove each of the following conjectures.

(a) $f(n) = O((f(n))^2)$.

(b) $f(n) = \Theta(f(n/2))$.

(a) Let $f(n) = \frac{1}{n}$

$$f(n) = O((f(n))^2) \rightarrow f(n) \leq C \times f(n)^2$$

$$\frac{1}{n} \leq C \times \frac{1}{n^2}$$

$\because C$ 為常數，無法平衡可以到 ∞ 的 n

$\therefore$ 無法找到一常數 C ，使 $f(n) = \frac{1}{n}$ 時 $f(n) = O(f(n)^2)$ *

(b) Let $f(n) = 2^n$

$$f(n) = \Theta(f(n/2)) \rightarrow C_1 \cdot f(n/2) \leq f(n) \leq C_2 \cdot f(n/2)$$

$$2^n \leq C_2 \cdot 2^{n/2}$$

$$2^{n/2} \leq C_2$$

$\because C_2$ 只是一個常數， n 可以向 ∞ 靠近

$\therefore$ 無法找到一常數 C_2 滿足 $f(n) = 2^n$ 時 $f(n) = \Theta(f(n/2))$ *